



An Assessment Report on
“Predicting Loan Default using Machine Learning”
submitted as partial fulfillment for the award of
**BACHELOR OF TECHNOLOGY
DEGREE**

SESSION 2024-25

in

INTRODUCTION TO AI

By

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Problem Statement:

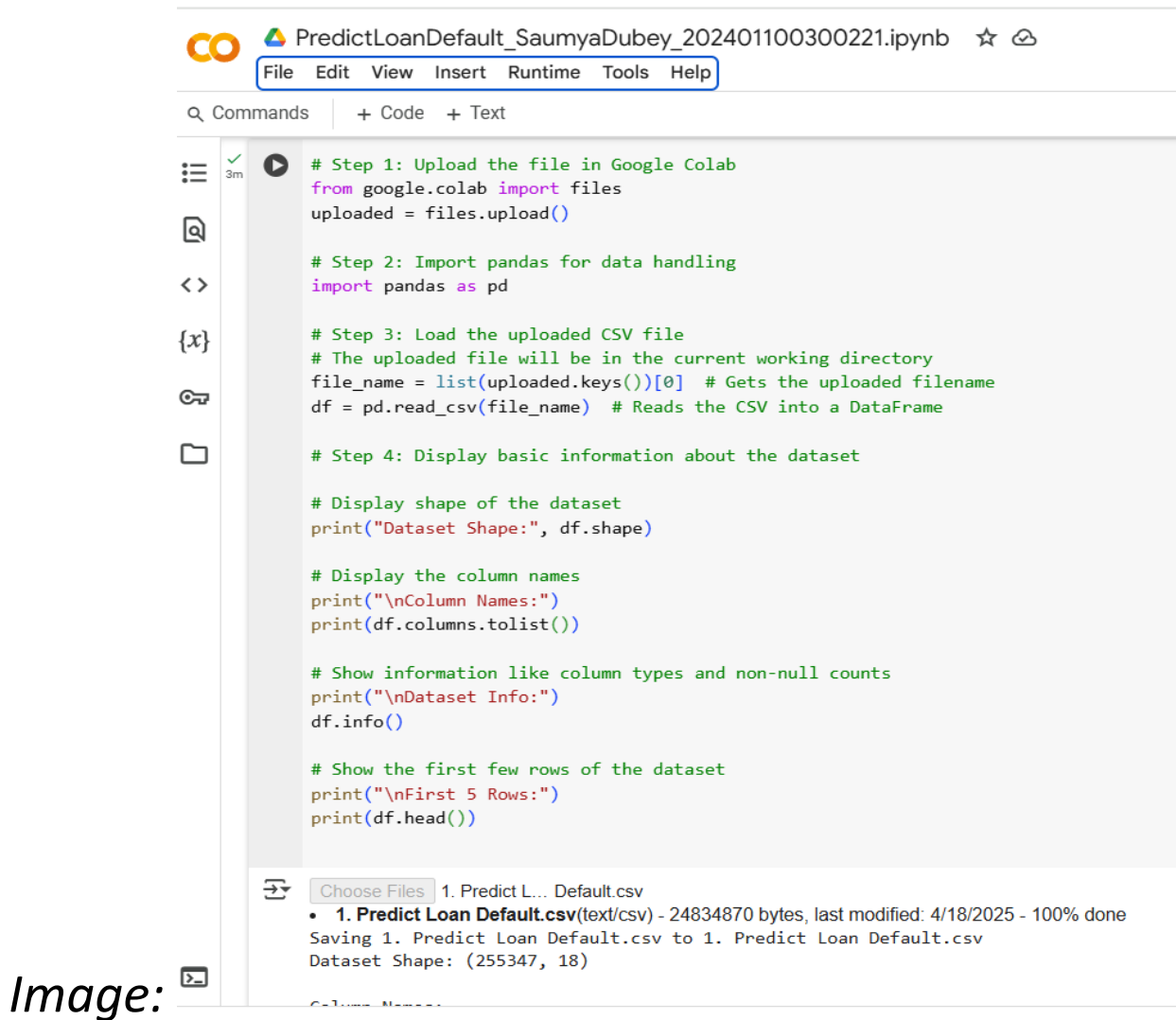
To develop a data-driven solution that predicts whether a loan applicant will default or not, using machine learning models on a real-world dataset.

Introduction

In the finance industry, loan default prediction is crucial to minimizing risk. Financial institutions aim to assess a borrower's risk profile before granting a loan. This project focuses on analyzing a dataset of loan applications and building a predictive model to classify whether a user will default.

The objective is to:

- Analyze the given loan dataset
- Clean and preprocess the data
- Apply machine learning techniques
- Evaluate the performance of the model



Methodology

1. **Data Loading:** Read the CSV dataset using Pandas.
2. **Data Inspection:** View data shape, columns, missing values.
3. **Data Preprocessing:**
 - Handle missing data
 - Encode categorical variables
 - Scale numerical values

4. **Model Building:** Apply classification algorithms like Logistic Regression, Decision Trees, or Random Forest.
 5. **Model Evaluation:** Use accuracy, precision, recall, and confusion matrix to evaluate performance.
-

Code

python

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```
# Upload CSV
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
# Import necessary libraries
```

```
import pandas as pd
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import classification_report, confusion_matrix
```

```
# Load dataset
```

```
file_name = list(uploaded.keys())[0]
```

```
df = pd.read_csv(file_name)
```

```
# Basic info

print(df.shape)

print(df.info())

print(df.head())


# Drop rows with null values

df.dropna(inplace=True)


# Encode categorical columns

df_encoded = pd.get_dummies(df, drop_first=True)


# Split features and label

X = df_encoded.drop("default", axis=1) # assuming 'default' is the target
y = df_encoded["default"]


# Train-test split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)


# Model training

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)


# Predictions and evaluation

y_pred = model.predict(X_test)

print("Classification Report:\n", classification_report(y_test, y_pred))
```

```
# Confusion matrix

sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d')

plt.title("Confusion Matrix")

plt.xlabel("Predicted")

plt.ylabel("Actual")


plt.show()
```

Output/Result

```

#      Column      Non-Null Count  Dtype
---  -
0      LoanID      255347 non-null    object
1      Age          255347 non-null    int64
2      Income       255347 non-null    int64
3      LoanAmount    255347 non-null    int64
4      CreditScore   255347 non-null    int64
5      MonthsEmployed 255347 non-null    int64
6      NumCreditLines 255347 non-null    int64
7      InterestRate   255347 non-null    float64
8      LoanTerm        255347 non-null    int64
9      DTIRatio        255347 non-null    float64
10     Education       255347 non-null    object
11     EmploymentType  255347 non-null    object
12     MaritalStatus   255347 non-null    object
13     HasMortgage     255347 non-null    object
14     HasDependents   255347 non-null    object
15     LoanPurpose     255347 non-null    object
16     HasCoSigner     255347 non-null    object
17     Default         255347 non-null    int64

dtypes: float64(2), int64(8), object(8)
memory usage: 35.1+ MB

First 5 Rows:
   LoanID  Age  Income  LoanAmount  CreditScore  MonthsEmployed \
0  I38PQUQ596  56  85994      50587          520           80
1  HP5K72WA7R  69  50432     124440          458           15
2  C10Z6DPJ8Y  46  84208     129188          451           26
3  V2KKSF3M3UN  32  31713     44799          743            0
4  EY08JDHTZP  60  20437      9139           633            8

   NumCreditLines  InterestRate  LoanTerm  DTIRatio  Education \
0                4          15.23         36      0.44  Bachelor's
1                1           4.81         60      0.68   Master's
2                3          21.17         24      0.31   Master's
3                3           7.07         24      0.23  High School
4                4           6.51         48      0.73  Bachelor's

```

```
PredictLoanDefault_SaumyaDubey_202401100300221.ipynb
File Edit View Insert Runtime Tools Help

12 maritalstatus 255347 non-null object
13 HasMortgage 255347 non-null object
14 HasDependents 255347 non-null object
15 LoanPurpose 255347 non-null object
16 HasCoSigner 255347 non-null object
17 Default 255347 non-null int64
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   NumCreditLines  InterestRate  LoanTerm  DTIRatio  Education  \
0                4          15.23         36        0.44  Bachelor's
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2                3          21.17         24        0.31  Master's
3                3           7.07         24        0.23  High School
4                4           6.51         48        0.73  Bachelor's

   EmploymentType  MaritalStatus  HasMortgage  HasDependents  LoanPurpose  \
0      Full-time      Divorced          Yes             Yes         Other
1      Full-time      Married           No             No          Other
2      Unemployed      Divorced          Yes             Yes          Auto
3      Full-time      Married           No             No         Business
4      Unemployed      Divorced           No             Yes          Auto

   HasCoSigner  Default
0           Yes         0
1           Yes         0
2           No         1
3           No         0
4           No         0

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```

References/Credits

- Dataset Source: Provided as 1. Predict Loan Default.csv
- Libraries Used: pandas, scikit-learn, matplotlib, seaborn
- Report structure reference: Assignment instructions (as per uploaded image)
- Coding in: Google Colab (Python)